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Browne

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(54) **UMBRELLA WITH PIVOTING LOCKING
DEVICE AND RELATED METHODS**

USPC 135/98, 20.1, 20.3, 21, 39
See application file for complete search history.

(71) Applicant: **ATLeisure, LLC**, Atlanta, GA (US)

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(72) Inventor: **William Martin Browne**, Atlanta, GA
(US)

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(73) Assignee: **ATLEISURE, LLC**, Atlanta, GA (US)

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A45B 23/00 (2006.01)

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2200/10

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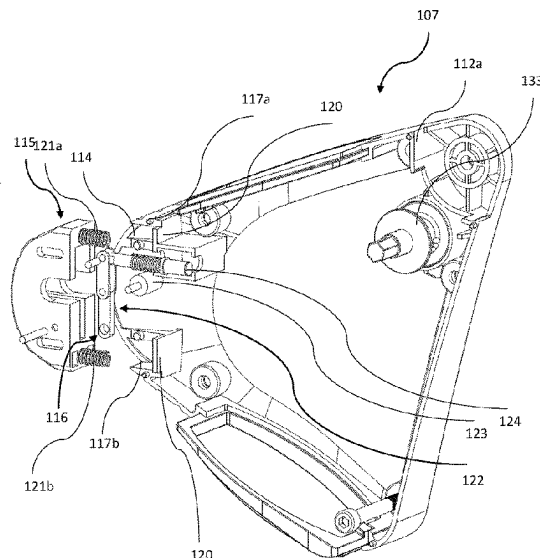
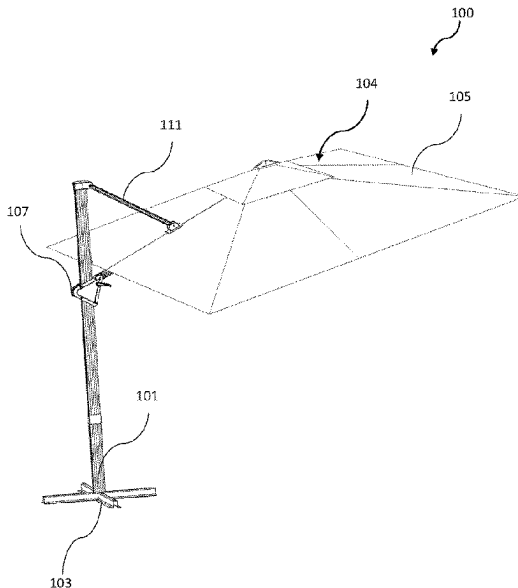
(74) *Attorney, Agent, or Firm* — Allen, Dyer, Doppelt +
Gilchrist, PA

(57)

ABSTRACT

An umbrella includes a vertical support having a track carried on an outer surface of the vertical support, the track extending longitudinally, and a canopy configured to switch between an open state and a closed state. The umbrella also includes a sliding collar coupled to the vertical support and configured to slidably engage the vertical support to control an angle of the canopy with respect to the vertical support, a first arm coupled between the sliding collar and the canopy, and a second arm coupled between the vertical support and the first arm. The sliding collar has a housing defining a recess, an activation button slidably received by the recess of the housing and switching between an extended state and a retracted state, and a locking device carried by the housing. The locking device is configured to pivot about the housing to selectively engage the track in the extended state.

20 Claims, 9 Drawing Sheets



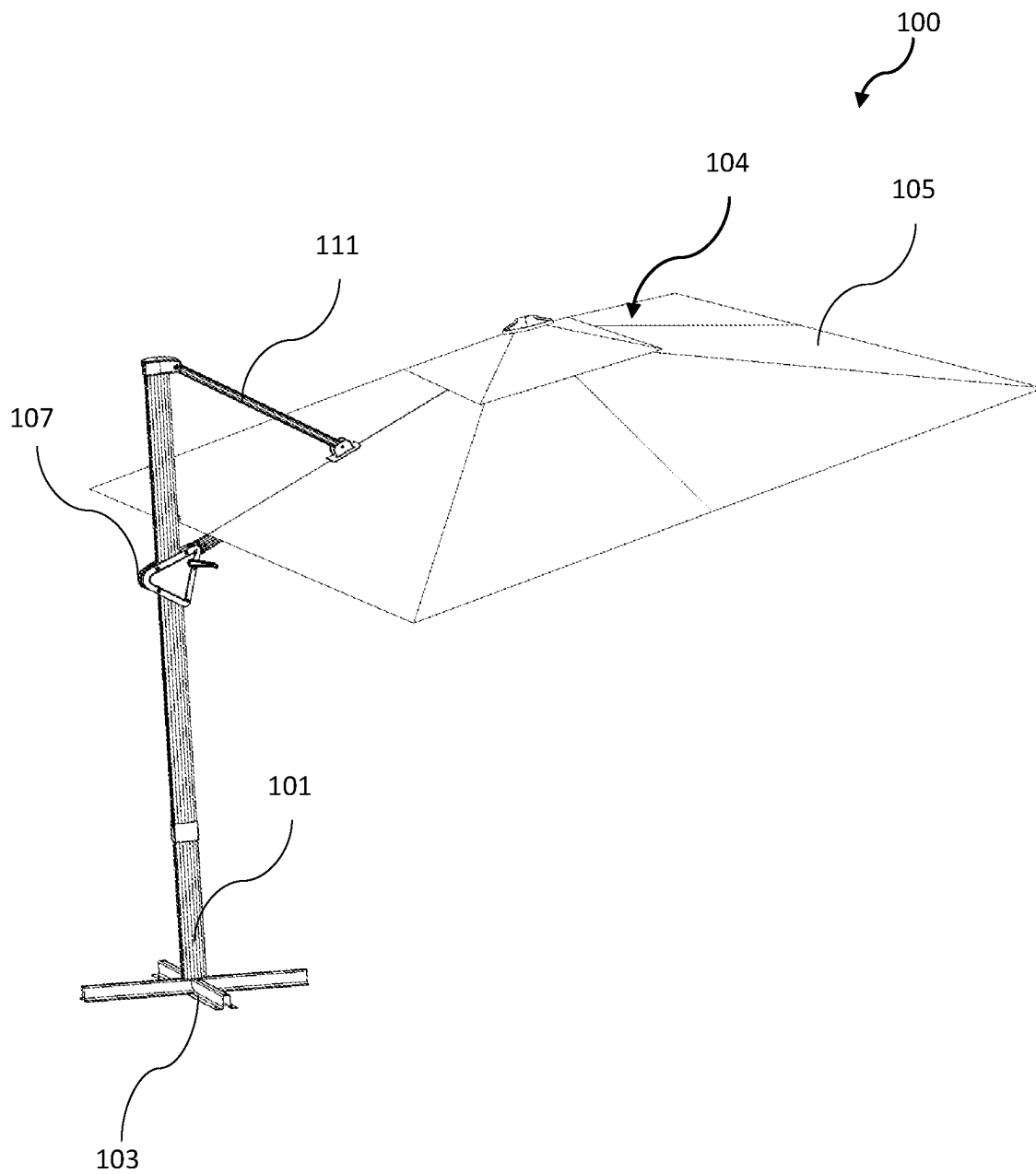
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**FIG. 1**

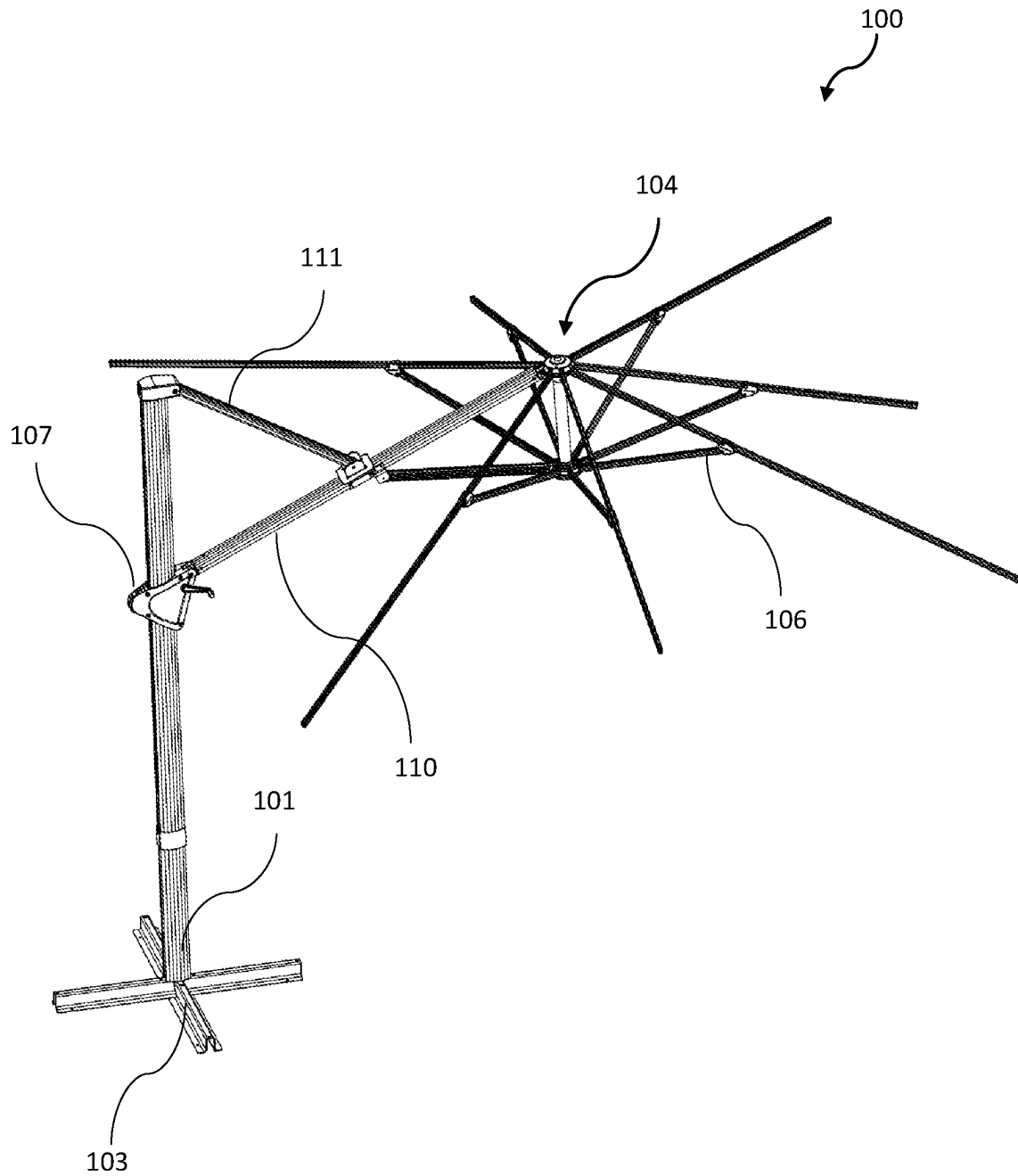


FIG. 2

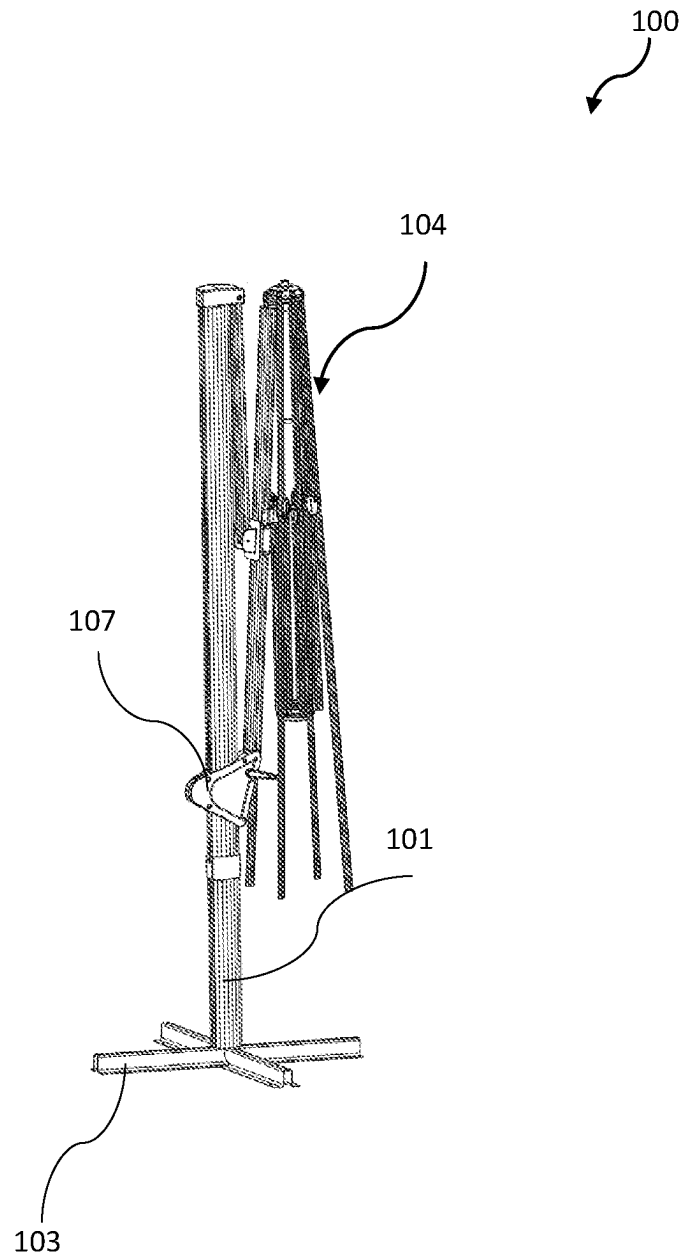


FIG. 3

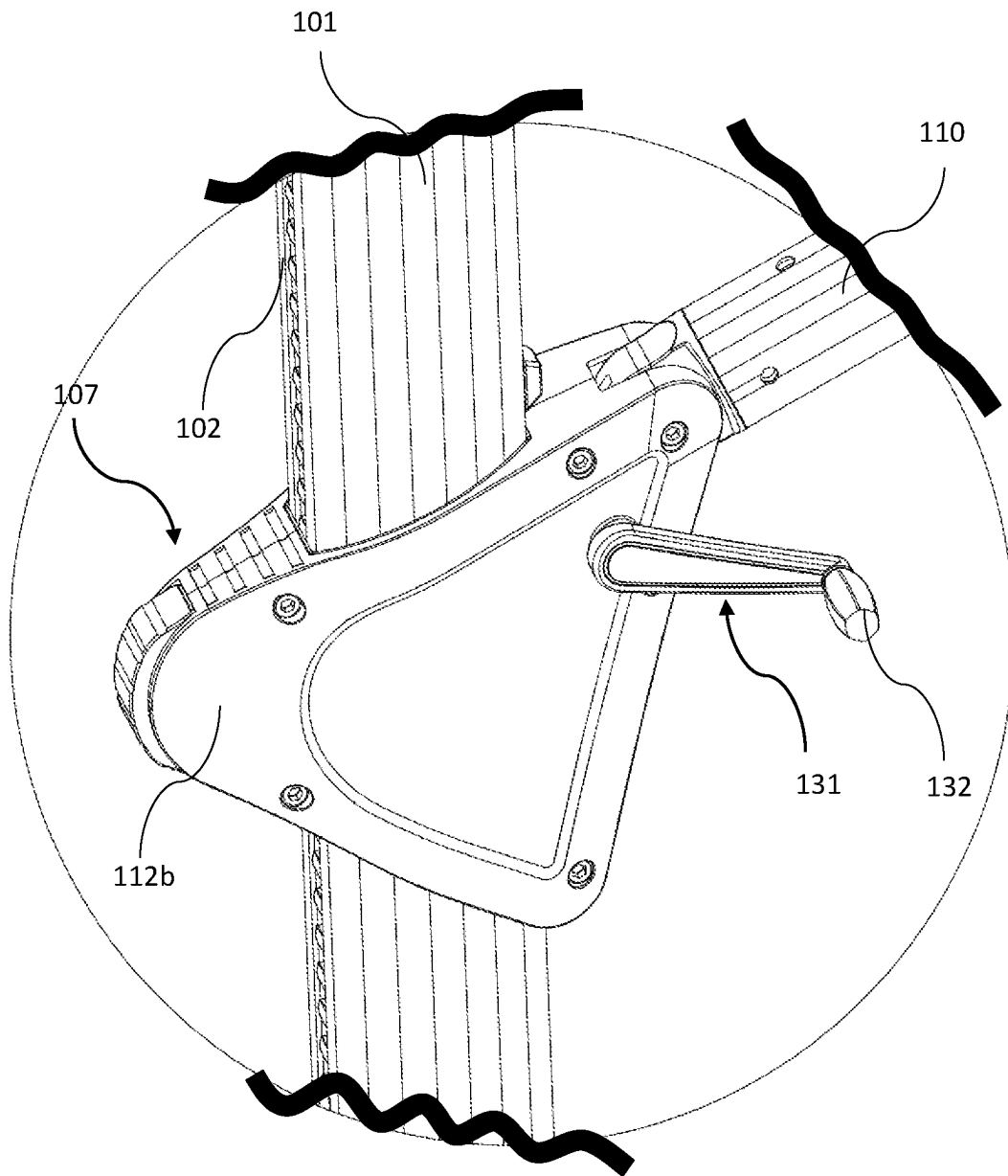


FIG. 4

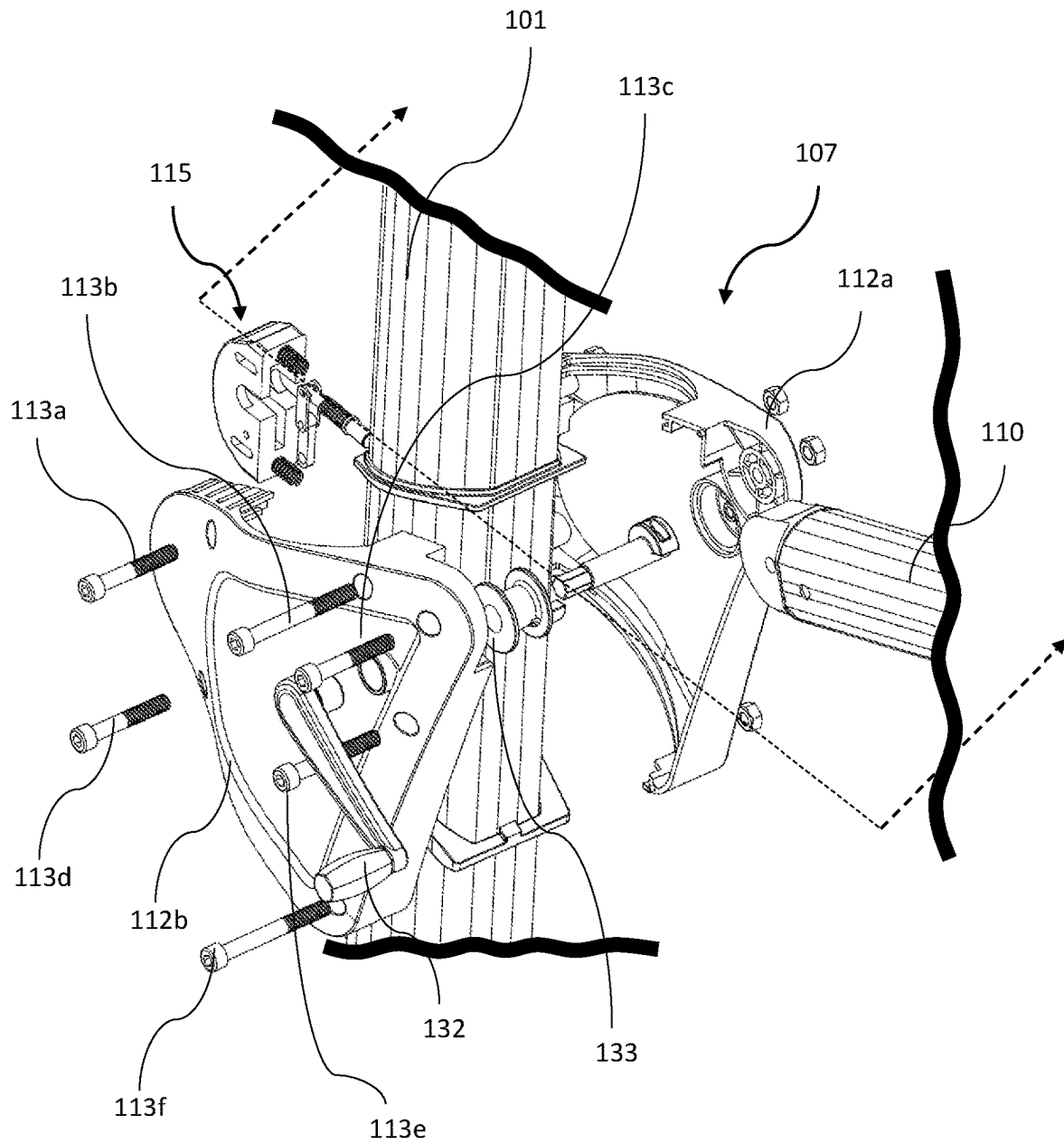
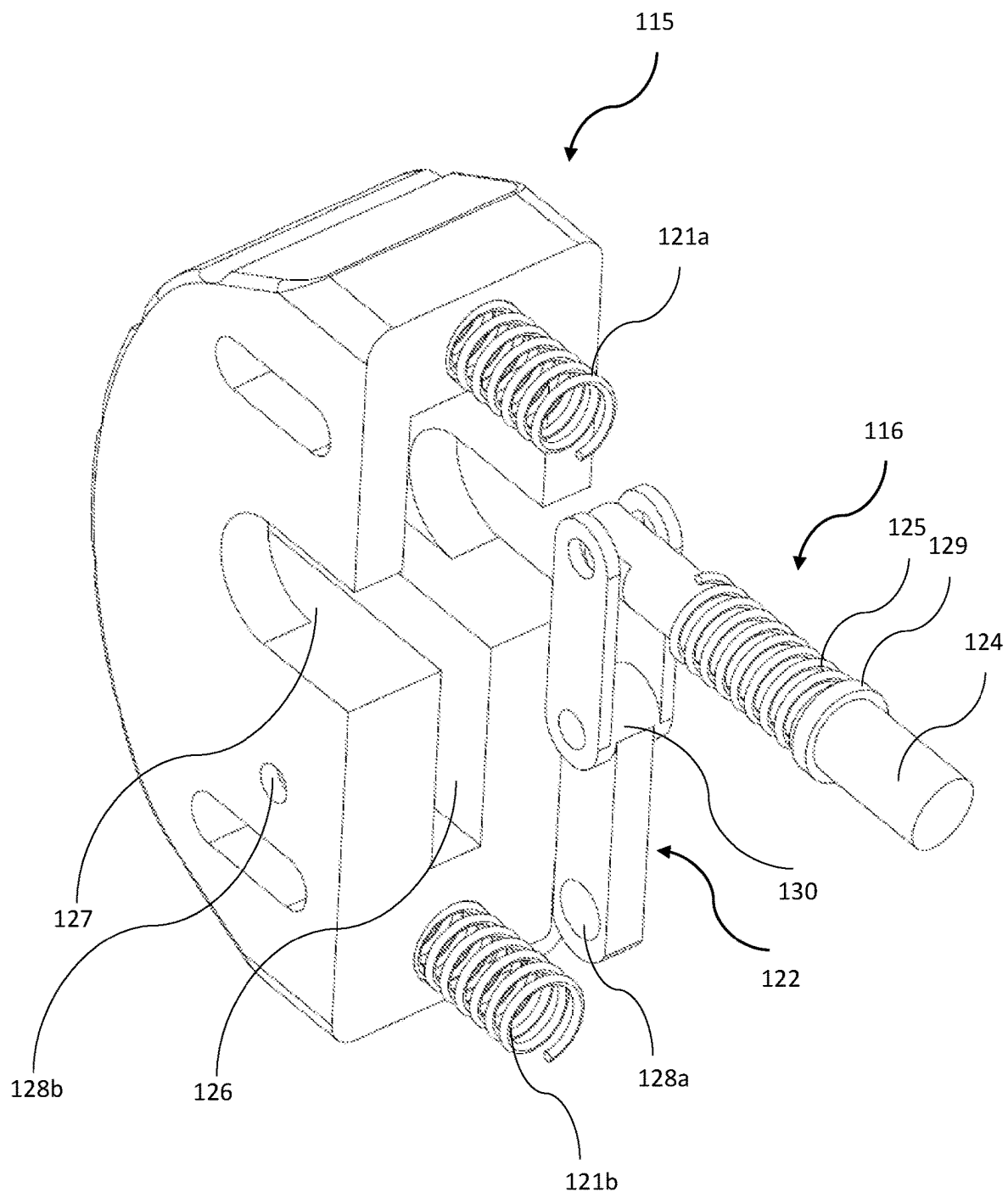


FIG. 5

**FIG. 6**

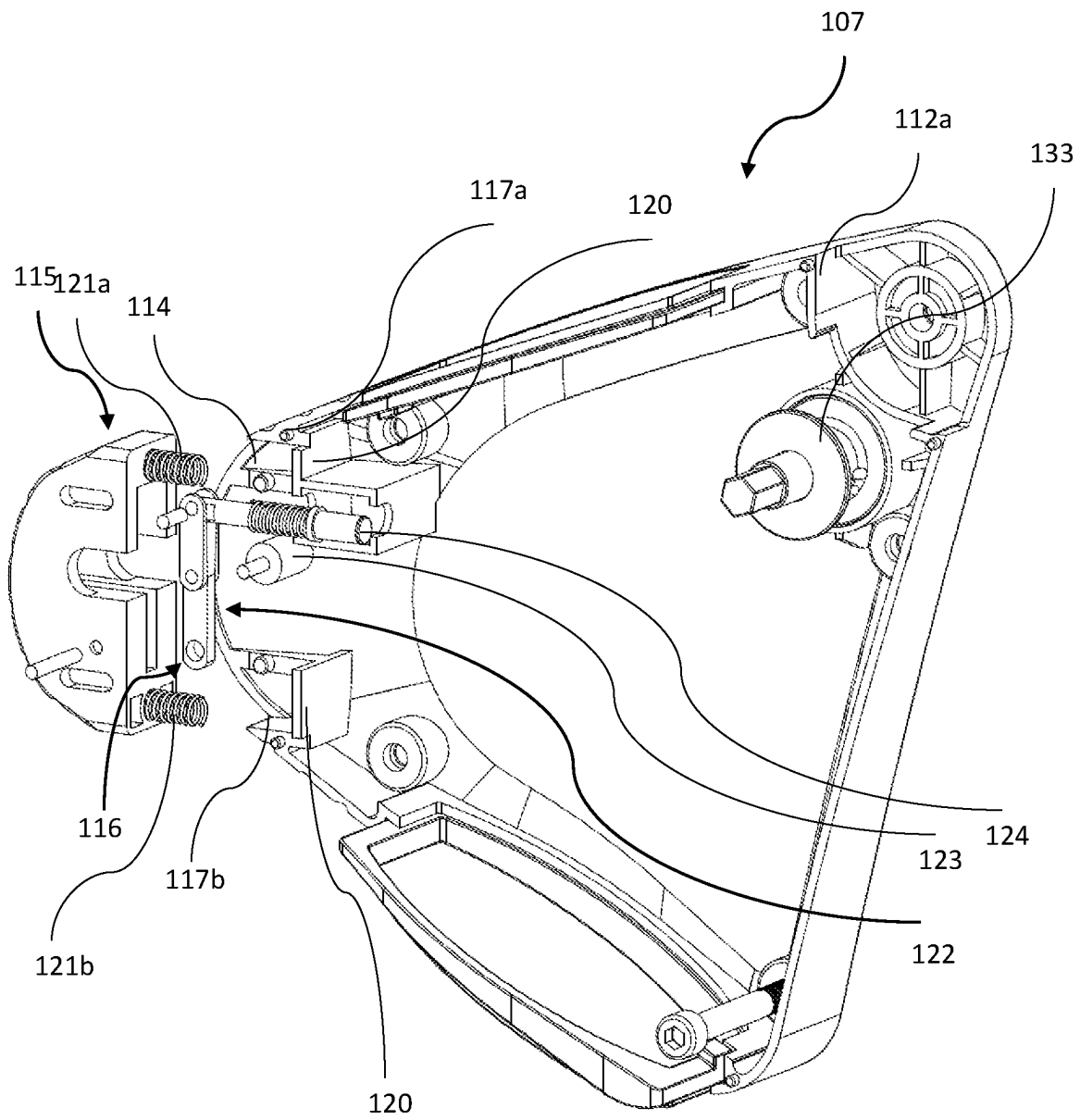


FIG. 7

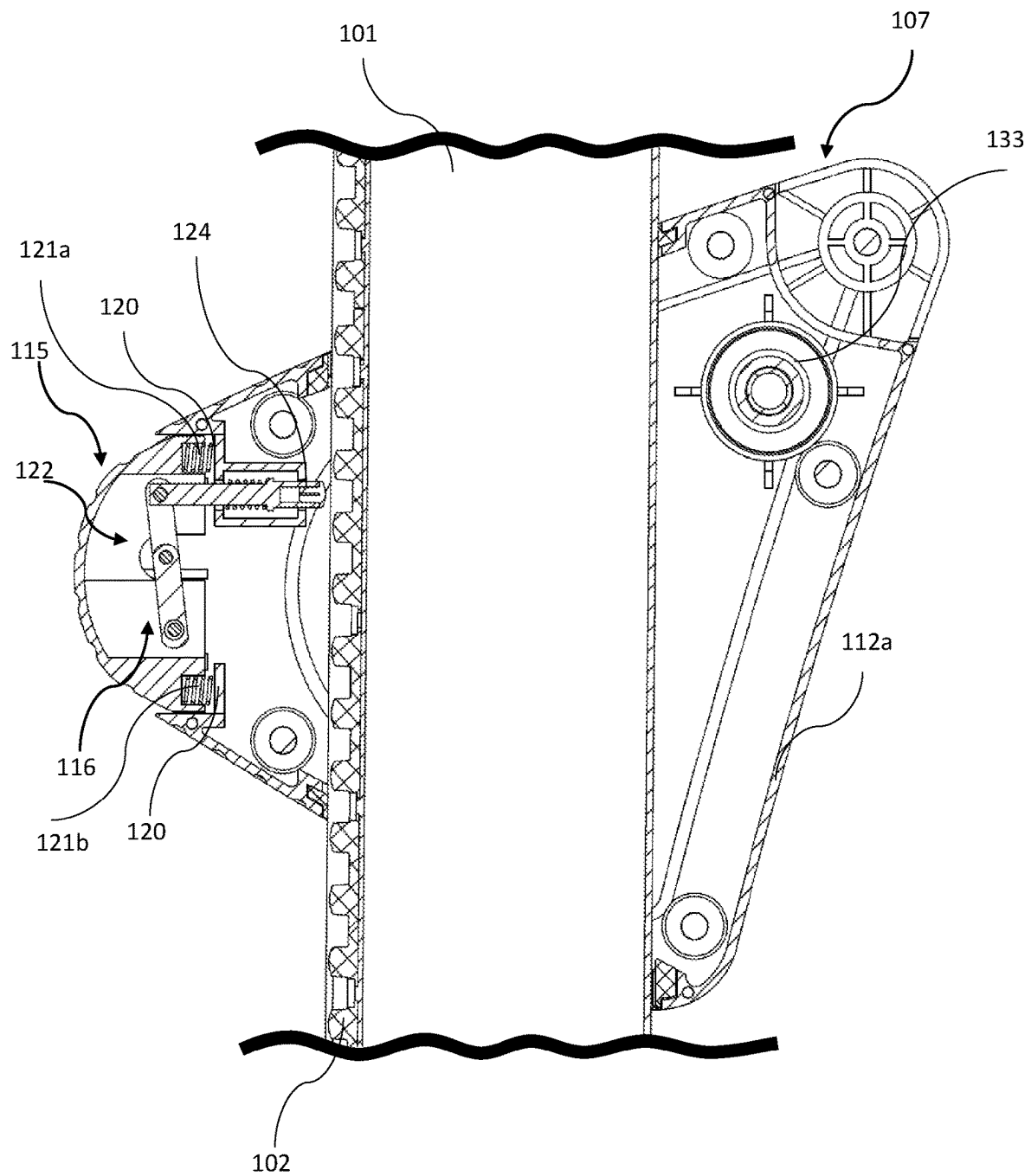


FIG. 8



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UMBRELLA WITH PIVOTING LOCKING DEVICE AND RELATED METHODS

RELATED APPLICATION

This application is based upon prior filed Application No. 63/385,883 filed Dec. 2, 2022, the entire subject matter of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to the field of outdoor accessories, and, more particularly, to an umbrella and related methods.

BACKGROUND

Large patio-style umbrellas are useful on patios, decks, pools, beaches, and in other outdoor settings for providing protection from the sun and allowing users a convenient way to enjoy shade while participating in outdoor activities. Typically, the umbrella comprises a mast extending from a base, and a canopy coupled to the mast opposite to the base. A drawback to this design is that the mast and base occupy some of the shaded area, thereby limiting the usable area.

In some approaches, as disclosed in, for example, U.S. Pat. No. 5,937,882 to Harbaugh, assigned to the present application's assignee, the umbrella uses an offset/cantilever arrangement. In this example, the main pole utilizes a cantilever arrangement to suspend the canopy to the side of the main pole so that the main pole does not occupy space in the shaded area beneath the canopy. For additional convenience, the angle of the canopy can be adjusted with respect to the main pole. A potential drawback to this design is the utilization of a crank mechanism both for raising and lowering the canopy and for adjusting the angle of the canopy. Typically, the crank mechanism is mounted to the main pole, and the rope moves along a complex path with many turning points. This arrangement may require more winding force, and the rope may bind and wear, adversely affecting the umbrella's operation.

In another approach disclosed in U.S. Pat. No. 8,104,492 to Dan, also assigned to the present application's assignee, the umbrella includes separate mechanisms for opening and closing the canopy and adjusting the angle of the canopy. Here, the sliding collar adjusts the angle of the canopy, and the winch on the sliding collar controls the opening and closing of the canopy.

SUMMARY

Generally, an umbrella comprises a vertical support having a track carried on an outer surface of the vertical support, the track extending longitudinally, and a canopy configured to switch between an open state and a closed state. The umbrella also includes a sliding collar coupled to the vertical support and configured to slidably engage the vertical support to control an angle of the canopy with respect to the vertical support, a first arm coupled between the sliding collar and the canopy, and a second arm coupled between the vertical support and the first arm. The sliding collar comprises a housing defining a recess, an activation button slidably received by the recess of the housing and switching between an extended state and a retracted state, and a locking device carried by the housing. The locking device is configured to pivot about the housing to selectively engage the track in the extended state.

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In particular, the recess may comprise a peripheral ledge, and the locking device may comprise first and second elastic members coupled between the peripheral ledge and the activation button and configured to bias the activation button in the extended state. The locking device may comprise a locking arm pivotably coupled to the housing and having a first end coupled to the activation button, and a second end opposite to the first end. The activation button may define a longitudinal recess therein to receive the locking arm. The locking device may comprise a pivot arm coupled between the housing and the locking arm. The activation button may define a medial recess therein to receive the pivot arm.

The locking device may comprise a locking pin coupled to the second end of the locking arm and configured to engage the track. The locking device may comprise a third elastic member carried by the locking pin and configured to bias the locking pin to engage the track. Also, the umbrella may also include a winding device carried by the sliding collar and coupled to the canopy, and the winding device may be configured to switch the canopy between the open state and the closed state.

Another aspect is directed to a method for making an umbrella. The method comprises forming a vertical support having a track carried on an outer surface of the vertical support, the track extending longitudinally, and coupling a sliding collar to the vertical support and configured to slidably engage the vertical support to control an angle of a canopy with respect to the vertical support. The canopy is configured to switch between an open state and a closed state. The method also comprises coupling a first arm between the sliding collar and the canopy, and coupling a second arm between the vertical support and the first arm. The sliding collar comprises a housing defining a recess, an activation button slidably received by the recess of the housing and switching between an extended state and a retracted state, and a locking device carried by the housing. The locking device is configured to pivot about the housing to selectively engage the track in the extended state.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of an umbrella in an open state, according to the present disclosure.

FIG. 2 is a perspective view of the umbrella from FIG. 1 without the canopy.

FIG. 3 is a perspective view of the umbrella from FIG. 1 in a closed state without the canopy.

FIG. 4 is a perspective view of the sliding collar from the umbrella from FIG. 1.

FIG. 5 is an exploded view of the sliding collar from the umbrella from FIG. 1.

FIG. 6 is a perspective view of the activation button and the locking device from the umbrella from FIG. 1.

FIG. 7 is a perspective view of the sliding collar from the umbrella from FIG. 1 with some of the collar housing removed.

FIG. 8 is a cross-sectional view of the sliding collar along line 8-8 from FIG. 5 with the locking arm not engaging the track and with the activation button in a retracted state.

FIG. 9 is a cross-sectional view of the sliding collar along line 8-8 from FIG. 5 with the locking arm engaging the track and with the activation button in an extended state.

DETAILED DESCRIPTION

The present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in

which several embodiments of the invention are shown. This present disclosure may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art. Like numbers refer to like elements throughout, and base **100** reference numerals are used to indicate similar elements in alternative embodiments.

Referring to FIGS. 1-9, an umbrella **100** according to the present disclosure is now described. The umbrella **100** may provide an approach to an issue with existing cantilever/offset umbrellas. In particular, the actuation of free movement of the sliding collar may be difficult for the user, making adjustment of the angle of the canopy difficult. In some approaches, the umbrella may use a compression mechanism to lock the sliding collar, but this may require greater than average hand strength, which may be less user friendly. Further, some users may not properly lock the sliding collar, permitting the canopy to unexpectedly move, for example, from a gust of wind.

The umbrella **100** comprises a vertical support **101** having a track **102** carried on an outer surface of the vertical support, and a base **103** coupled to a lowermost portion of the vertical support. Also, the vertical support **101** may comprise a metallic material, such as aluminum, but may comprise one or more metals in some embodiments. Indeed, any rigid material with sufficient mechanical strength may be used, for example, plastic, alternatively or in combination with the metallic material.

The track **102** extends longitudinally along the vertical support **101** and may extend over some or all of the vertical support. As perhaps best seen in FIGS. 8-9, the track **102** illustratively comprises interweaved teeth and recesses. In some embodiments, the track **102** may be integral with the vertical support **101**, and in other embodiments, the track may be modular and installed onto the vertical support.

The umbrella **100** comprises a canopy **104** configured to switch between an open state (FIG. 1) and a closed state (FIG. 3). The canopy **104** includes a fabric shell **105**, and a structure **106** supporting the fabric shell. As will be appreciated, the structure **106** comprises a plurality of support arms, and a collar coupled to the plurality of support arms. The umbrella **100** also includes a sliding collar **107** coupled to the vertical support **101** and configured to slidably engage the vertical support to control an angle of the canopy **104** with respect to the vertical support. As shown in FIG. 3, adjusting the sliding collar **107** downward stows the canopy **104** by decreasing the angle with the vertical support **101**. As shown in FIGS. 1-2, adjusting the sliding collar **107** upward extends the canopy **104** outward by increasing the angle with the vertical support **101**.

The umbrella **100** also includes a first arm **110** pivotally coupled between the sliding collar **107** and the canopy **104**. The umbrella **100** includes a second arm **111** pivotally coupled between an uppermost portion of the vertical support **101** and the first arm **110**.

The sliding collar **107** illustratively comprises first and second housing segments **112a-112b**, and a plurality of fasteners **113a-113f** coupling together the first and second housing segments. Each of the plurality of fasteners **113a-113f** illustratively comprises a threaded machined screw, and an opposing threaded nut. Of course, other forms of fastening can be used, such as an adhesive bonding. The first and second housing segments **112a-112b** cooperate to define a recess **114**.

The sliding collar **107** comprises an activation button **115** slidably received by the recess **114** of the first and second housing segments **112a-112b** and switching between an extended state (FIG. 9) and a retracted state (FIG. 8). The sliding collar **107** comprises a locking device **116** carried by the first and second housing segments **112a-112b**.

As perhaps best seen in FIG. 7, the recess **114** illustratively comprises first and second inner walls **117a-117b** extending inwardly from an outer surface of the first housing segment **112a**, and a peripheral ledge **120** extending from an innermost end of the first and second inner walls. The locking device **116** illustratively comprises first and second elastic members **121a-121b** coupled between the peripheral ledge **120** and the activation button **115** and configured to bias the activation button in the extended state. In the illustrated example, each of the first and second elastic members **121a-121b** comprises a coil spring.

The locking device **116** illustratively includes a locking arm **122** pivotally coupled to the first housing segment **112a** and having a first end coupled to the activation button **115**, and a second end opposite to the first end. In the illustrated embodiment, the first end of the locking arm **122** defines a passageway **128a** to be aligned with another passageway **128b** on the activation button **115**, both of which receiving a fastener, such as an elongate pin. The locking device **116** comprises a pivot arm **123** (FIG. 7) coupled between the first housing segment **112a** and a medial portion of the locking arm **122**.

The locking device **116** comprises a locking pin **124** coupled to the second end of the locking arm **122** and configured to engage the track **102**. The locking device **116** comprises a third elastic member **125**, for example, the illustrated coil spring, (FIG. 6) carried by the locking pin **124** and configured to bias the locking pin to engage the track **102**. In particular, the locking pin **124** illustratively comprises a flanged rim **129** to retain the third elastic member **125** against adjacent portions of the first housing segment **112a** (i.e., portions opposite the peripheral ledge **120** best seen in FIGS. 8-9).

As perhaps best seen in FIG. 6, the activation button **115** defines a longitudinal recess **126** therein to receive the locking arm **122**. The activation button **115** also defines a medial recess **127** therein to receive the pivot arm **123**. In this illustrated example, the locking arm **122** comprises first and second arm segments coupled to the locking pin **124**, a third arm segment, and an arm junction **130** coupling the first and second arm segment with the third arm segment. The arm junction **130** also receives the pivot arm **123**.

As will be appreciated and perhaps best seen in FIG. 8, the first end of the locking arm **122** is coupled to the activation button **115**, and when the activation button is depressed into the retracted state, the first end of the locking arm **122** is pivoted downward, causing the second end of the locking arm and the locking pin **124** to travel upward and away from the track **102**. In this retracted state, the first, second, and third elastic members **121a-121b**, **125** are in a compressed state. As seen in FIG. 9, when the activation button **115** is released, due to the outward biasing of the first and second elastic members **121a-121b**, the activation button returns to the extended state, which causes the first end of the locking arm **122** to pivot upward, causing the second end of the locking arm and the locking pin **124** to travel downward to the track **102** (aided by the downward biasing of the third elastic member **125**). In other words, via these components, the locking device **116** is configured to pivot about the first housing segment **112a** to selectively engage the track **102** in the extended state.

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Advantageously, this makes adjustment of the sliding collar **107** easier than with existing approaches. Indeed, to adjust the angle of the canopy **104**, the user need only depress the activation button **115** and gently slide the sliding collar **107**. When the sliding collar **107** has reached the desired position (i.e., the canopy **104** has the desired angle with respect to the vertical support **101**), the user releases the activation button **115**, which springs into the extended position, locking the sliding collar **107** into place. In contrast to existing approaches, the sliding collar **107** is reliably locked into place without significant user effort.

As seen in FIGS. 4-5, the umbrella **100** illustratively comprises a winding device **131** carried by the sliding collar **107** and coupled to the canopy **104**. The winding device **131** comprises a hand crank **132**, and a hub **133** coupled to the hand crank and receiving a cable. The cable is routed through the first arm **110** and to the canopy **104**. The winding device **131** is configured to switch the canopy **104** between the open state and the closed state using the hand crank **132**.

Another aspect is directed to a method for making an umbrella **100**. The method comprises forming a vertical support **101** having a track **102** carried on an outer surface of the vertical support. The track **102** extends longitudinally. The method also includes coupling a sliding collar **107** to the vertical support **101** and configured to slidably engage the vertical support to control an angle of a canopy **104** with respect to the vertical support. The canopy **104** is configured to switch between an open state and a closed state. The method also comprises coupling a first arm **110** between the sliding collar **107** and the canopy **104**, and coupling a second arm **111** between the vertical support **101** and the first arm. The sliding collar **107** comprises first and second housing segments **112a-112b** defining a recess **114**, an activation button **115** slidably received by the recess of the housing and switching between an extended state and a retracted state, and a locking device **116** carried by the first and second housing segments **112a-112b**, the locking device configured to pivot about the first and second housing segments to selectively engage the track **102** in the extended state.

Other features relating to umbrellas are disclosed in U.S. Pat. Nos. 7,604,015; 8,104,492; 9,470,413; 7,882,846; 9,504,298; 7,533,679; 9,622,569; 5,937,882; 6,889,953; and 9,655,417; and pending U.S. patent application Ser. No. 18/186,375, all incorporated herein by reference in their entirety.

Many modifications and other embodiments of the present disclosure will come to the mind of one skilled in the art having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is understood that the present disclosure is not to be limited to the specific embodiments disclosed, and that modifications and embodiments are intended to be included within the scope of the appended claims.

The invention claimed is:

1. An umbrella comprising:

a vertical support having a track carried on an outer surface of the vertical support, the track extending longitudinally;

a canopy configured to switch between an open state and a closed state;

a sliding collar coupled to the vertical support and configured to slidably engage the vertical support to control an angle of the canopy with respect to the vertical support;

a first arm coupled between the sliding collar and the canopy; and

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a second arm coupled between the vertical support and the first arm;

the sliding collar comprising

a housing defining a recess,

an activation button received by the recess of the housing and switching between an extended state and a retracted state, and

a locking device carried by the housing, the locking device comprising

a locking arm received in the activation button and pivotably coupled to the housing and having a first end coupled to the activation button, and a second end opposite to the first end, and

a locking pin coupled to the second end of the locking arm and configured to selectively engage the track in the extended state.

2. The umbrella of claim 1 wherein the recess comprises a peripheral ledge; and wherein the locking device comprises first and second elastic members coupled between the peripheral ledge and the activation button and configured to bias the activation button in the extended state.

3. The umbrella of claim 1 wherein the activation button defines a longitudinal recess therein to receive the locking arm.

4. The umbrella of claim 1 wherein the locking device comprises a pivot arm coupled between the housing and the locking arm.

5. The umbrella of claim 4 wherein the activation button defines a medial recess therein to receive the pivot arm.

6. The umbrella of claim 1 wherein the locking device comprises an elastic member carried by the locking pin and configured to bias the locking pin to engage the track.

7. The umbrella of claim 1 further comprising a winding device carried by the sliding collar and coupled to the canopy; and wherein the winding device is configured to switch the canopy between the open state and the closed state.

8. An umbrella comprising:

a vertical support having a track carried on an outer surface of the vertical support, the track extending longitudinally;

a canopy configured to switch between an open state and a closed state;

a sliding collar coupled to the vertical support and configured to slidably engage the vertical support to control an angle of the canopy with respect to the vertical support;

a winding device carried by the sliding collar and coupled to the canopy, the winding device configured to switch the canopy between the open state and the closed state;

a first arm coupled between the sliding collar and the canopy; and

a second arm coupled between the vertical support and the first arm;

the sliding collar comprising

a housing defining a recess having a peripheral ledge, an activation button received by the recess of the housing and switching between an extended state and a retracted state, and

a locking device carried by the housing, the locking device comprising

first and second elastic members coupled between the peripheral ledge and the activation button and configured to bias the activation button in the extended state,

a locking arm received in the activation button and pivotably coupled to the housing and having a first

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end coupled to the activation button, and a second end opposite to the first end, and
a locking pin coupled to the second end of the locking arm and configured to selectively engage the track in the extended state.

9. The umbrella of claim 8 wherein the activation button defines a longitudinal recess therein to receive the locking arm.

10. The umbrella of claim 8 wherein the locking device comprises a pivot arm coupled between the housing and the locking arm.

11. The umbrella of claim 10 wherein the activation button defines a medial recess therein to receive the pivot arm.

12. The umbrella of claim 8 wherein the locking device comprises a locking pin coupled to the second end of the locking arm and configured to engage the track.

13. The umbrella of claim 12 wherein the locking device comprises a third elastic member carried by the locking pin and configured to bias the locking pin to engage the track.

14. A method for making an umbrella, the method comprising:

forming a vertical support having a track carried on an outer surface of the vertical support, the track extending longitudinally;

coupling a sliding collar to the vertical support and configured to slidably engage the vertical support to control an angle of a canopy with respect to the vertical support, the canopy configured to switch between an open state and a closed state;

coupling a first arm between the sliding collar and the canopy; and

coupling a second arm between the vertical support and the first arm;

the sliding collar comprising

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a housing defining a recess,
an activation button received by the recess of the housing and switching between an extended state and a retracted state, and

a locking device carried by the housing, the locking device comprising

a locking arm received in the activation button and pivotably coupled to the housing and having a first end coupled to the activation button, and a second end opposite to the first end, and

a locking pin coupled to the second end of the locking arm and configured to selectively engage the track in the extended state.

15. The method of claim 14 wherein the recess comprises a peripheral ledge; and wherein the locking device comprises first and second elastic members coupled between the peripheral ledge and the activation button and configured to bias the activation button in the extended state.

16. The method of claim 14 wherein the activation button defines a longitudinal recess therein to receive the locking arm.

17. The method of claim 14 wherein the locking device comprises a pivot arm coupled between the housing and the locking arm.

18. The method of claim 17 wherein the activation button defines a medial recess therein to receive the pivot arm.

19. The method of claim 14 wherein the locking device comprises an elastic member carried by the locking pin and configured to bias the locking pin to engage the track.

20. The method of claim 14 further comprising coupling a winding device to be carried by the sliding collar and to be coupled to the canopy; and wherein the winding device is configured to switch the canopy between the open state and the closed state.

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